

Biomedical Data Processing — Final Project

Retake report: ECG-based identification and EEG-based sleep staging

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Statement. This retake was carried out individually. The starting point was the implementation submitted in the first exam period; every change made since is my own and is marked in the code with an inline `# RETAKE CHANGE` comment. Section 3 lists those changes and maps each one to the section of this report that justifies it.

Deliverables. `Part 1/2025_project_part1.ipynb` and `Part 2/2025_project_part2.ipynb`. Both notebooks run top to bottom and write every figure and table used here into `report/figures`, so no number in this document is typed by hand. Part 1 executes in about 30 s. Part 2 caches its three expensive steps (the ICA fit, the extracted features and the cross-validation table) in `Part 2/cache/`; with a cold cache it takes roughly 25 minutes, with a warm cache about 40 s.

1 Part I — ECG-based identification

1.1 Pipeline

The recording is a single ECG channel, 605 s at 200 Hz (121 006 samples), containing the heartbeats of five subjects in random order. The identification chain is: R-peak detection with Pan–Tompkins → estimation of an average PQRST template and of a noise-only signal → Wiener filtering → re-detection and segmentation into 0.7 s beats → `db4` wavelet decomposition → K -means with $K = 5$, optionally after PCA. The clustering itself never sees the labels; they are used only to score a finished partition, through the permutation search in `match_labels`.

1.2 Answers to the overview-sheet questions

Which noise sources are present in the ECG?

Three, all of them stationary, and all three visible in the Welch spectrum of the raw recording (Fig. 1).

- **Baseline wander below 0.7 Hz**, carrying 22.3 % of the total power of the record. Each subject presses on the sensor with a different pressure, so the electrode–skin contact potential shifts every time a new person touches the electrode.
- **Powerline interference at exactly 50 Hz**, standing 15.1 dB above the local noise floor. There is no line at 60 Hz and no harmonic at 100 Hz, which is consistent with a European recording site (the data were recorded in Berlin).
- **Broadband muscle and contact noise above roughly 60 Hz**, from small finger and forearm contractions while the subject holds the sensor.

Change: in the first exam period these three sources were named from the time-domain trace alone. The question asks *why*, so the answer is now backed by a measurement.

How did you compute the Wiener filter? What methods did you use for the PSD estimation?

The filter is the frequency-domain Wiener solution

$$W(f) = \frac{S_d(f)}{S_d(f) + S_n(f)},$$

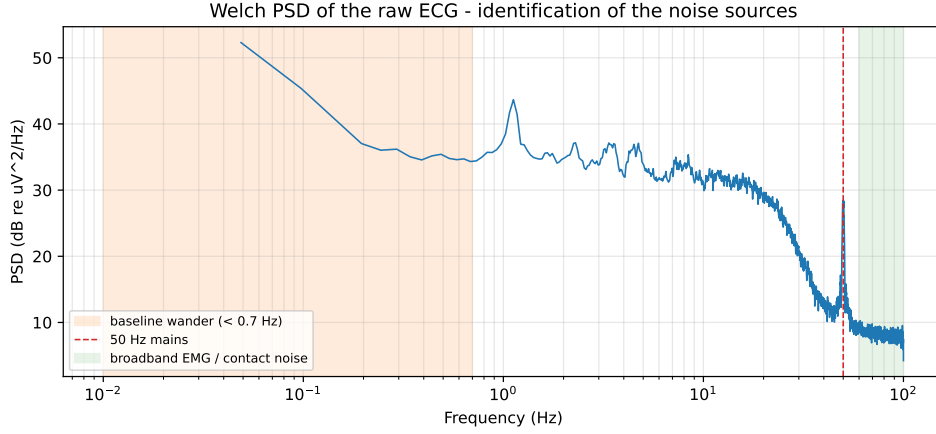


Figure 1: Welch PSD of the raw ECG (Hann window, 4096 samples, 50 % overlap, $\Delta f = 0.049 \text{ Hz}$). The resolution is deliberately fine so that the narrow mains line can be separated from the broadband ECG spectrum.

built on an L -point grid and applied to the full record with weighted overlap-add (`wola_filt`). S_d and S_n are estimated *differently*, because they describe different kinds of object.

Noise, S_n : Welch. The noise model is the concatenation of all samples that lie more than 0.4s after one R-peak and more than 0.3s before the next. It is long (≈ 4 minutes) and stationary, so Welch’s method is appropriate: Hann window, `nperseg` = $L = 512$, 50 % overlap, `nfft` = L , two-sided density scaling. A raw periodogram of a random process has ≈ 100 % relative standard deviation regardless of record length; only averaging — here over about 90 overlapping segments — makes the estimate usable. The Hann taper suppresses leakage from the strong low-frequency components into the rest of the band.

Desired signal, S_d : periodogram of the zero-padded template. The averaged PQRST complex is a single 140-sample *deterministic transient*, not a realisation of a stationary process. There is nothing to average over, so Welch is the wrong tool. S_d is the periodogram of the mean-removed template zero-padded to L , normalised by $f_s N_d$ with N_d the true template length so that its units match those of S_n . Zero-padding only interpolates the frequency axis onto the grid the filter uses; it adds no information and distorts nothing.

Change: previously S_d was computed with `welch(nperseg=128, noverlap=64)` on a 140-sample template. That configuration admits exactly one segment — the second would need samples 64–191, which do not exist — so the call silently returned a Hann-windowed periodogram of samples 0–127 and discarded the last 12, while being described in the overview sheet as an average over overlapping segments.

Filter length. $L = 512$ (2.56 s, $\Delta f = 0.391 \text{ Hz}$), chosen by the sweep in §1.3.2 rather than assumed. The mean is removed from both the template and the noise before the PSDs are taken: a DC offset is an artefact of the electrode contact rather than a feature of the PQRST complex, it would place a large spike at $f = 0$ in S_d and bias the filter towards passing the very baseline wander we are removing, and Wiener theory assumes a zero-mean noise process.

Did you normalise the wavelet coefficients before classification (without dimensionality reduction)?

Yes — each of the 173 coefficients is z -scored across segments before K -means. K -means minimises a Euclidean criterion, so every dimension contributes to the distance in proportion to its numerical scale rather than to its information content. The wavelet sub-bands are on very different scales here: the `ca5` coefficients track the residual baseline level and are an order of magnitude larger

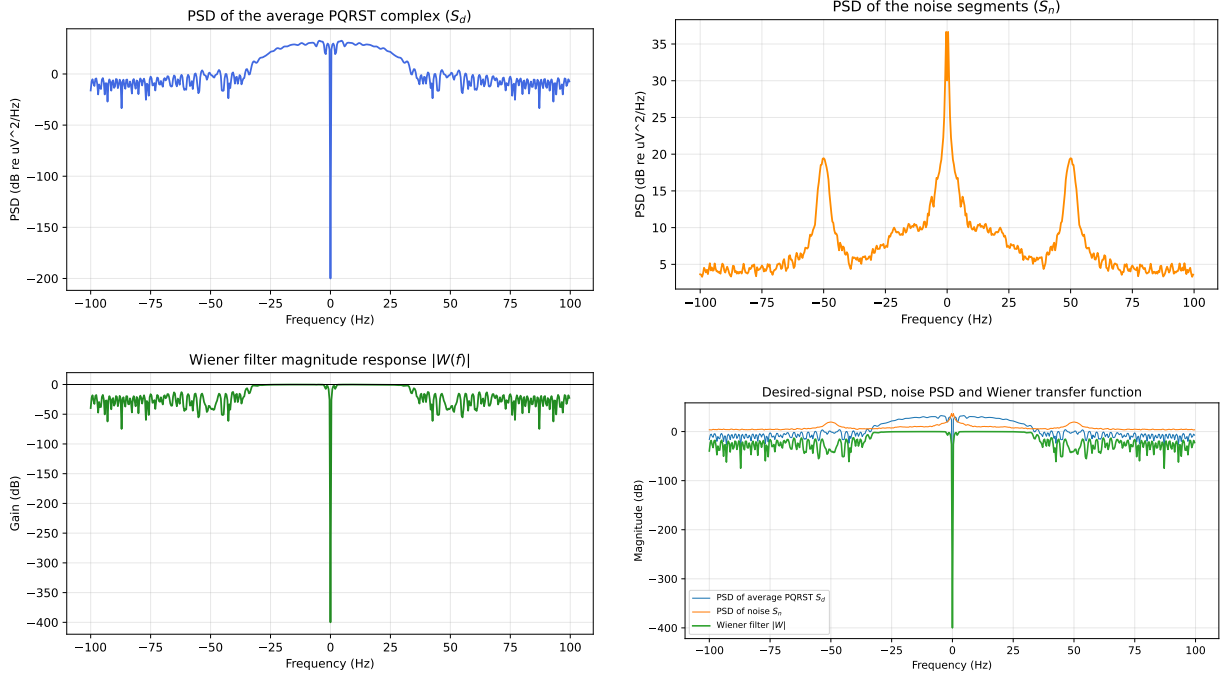


Figure 2: Deliverable plots for the Wiener filter. PSDs are plotted as $10 \log_{10} S$ because a PSD is a power density; the transfer function is plotted as $20 \log_{10} |W|$ because it is an amplitude gain. Change: the two stand-alone PSD plots previously used $20 \log_{10}$, which doubled their dB axis and contradicted the combined plot.

than the fine-scale cD1 coefficients. Without standardisation the partition is driven by the baseline offset of the recording session; with it, by the beat morphology. The effect is not marginal: **0.489 without standardisation against 1.000 with it** (Table 5).

How did you reduce the feature dimensionality as much as possible? Did you use any normalisation? If yes, when?

PCA on the standardised 173-dimensional coefficient matrix, keeping $N = 8$ components (54.9 % of the variance). This is a 95 % reduction in dimensionality for a loss of 0.004 in accuracy ($1.000 \rightarrow 0.9956$).

Normalisation before PCA: yes. PCA maximises variance along its components and is therefore scale-dependent by construction; on unstandardised coefficients PC1 would simply point along the baseline offset regardless of whether that offset says anything about who the subject is.

Normalisation after PCA: no. The components come out already decorrelated and ordered by decreasing variance, and that ordering is informative. Re-standardising them would rescale every component to unit variance and so promote the trailing, mostly-noise directions until they weigh as much in the K -means distance as the leading ones. Measured over ten seeds, post-PCA scaling is equal or worse at every N tested; at $N = 8$ it costs $0.9956 \rightarrow 0.9882$.

How N was chosen. Not by a variance threshold. PCA is unsupervised and maximises variance, not class separability, so an explained-variance rule is only a proxy for what we care about — and here the two disagree sharply: the 90 % rule would keep **40** components, whereas accuracy has already saturated by $N \approx 8$ (Fig. 4a). $N = 8$ is the elbow of the measured curve.

Table 1: Accuracy before and after dimensionality reduction (mean over ten K -means seeds).

Feature space	Dimensions	Accuracy
Wavelet coefficients, z -scored	173	1.0000
PCA of the same	8	0.9956

1.3 Design choices not covered by the overview sheet

1.3.1 Pan–Tompkins parameters, and how the detector was validated

The detector follows the standard chain: 5 Hz–11 Hz band-pass (two 4th-order Butterworth sections applied with `filtfilt`), the five-point derivative $y[n] = \frac{1}{8}(2x[n] + x[n-1] - x[n-3] - 2x[n-4])$, squaring, a 150 ms moving-window integrator, peak picking with a minimum inter-beat distance of 250 ms (equivalent to 240 beats per minute, comfortably above any physiological rate), and a ± 100 ms local search that moves each detection onto the true R-peak. Because `filtfilt` is zero-phase, the band-pass introduces no group delay to compensate; the derivative is applied causally with `lfilter`, and the local search absorbs its delay.

The threshold is a fixed multiple of the mean of the integrated signal rather than the adaptive dual-threshold scheme with search-back of the original publication. This is a simplification, and it is defensible here only because the recording is short, the noise is stationary, and — as shown next — it recovers the right number of beats.

Validation. Each subject only ever measured a single beat at a time, so every run of constant ground-truth label corresponds to one measurement episode. Dividing each run by the median run length (0.94 s) converts the label track into an independent estimate of the true beat count: **approximately 675 heartbeats**. The detector finds 679 peaks on the raw signal and 676 after Wiener filtering, and the resulting per-subject shares (Table 2) match the label track to within half a percentage point. This estimate is used only to validate the detector and never inside the algorithm.

Table 2: Share of segments per subject, and the independent estimate from the label track.

Subject	1	2	3	4	5
Segments	117	147	40	115	257
Share (%)	17.31	21.75	5.92	17.01	38.02
Estimate from labels (%)	17.4	21.3	5.9	17.3	38.1

1.3.2 Wiener filter length

L sets two things at once: the frequency resolution $\Delta f = f_s/L$ and the length of the WOLA analysis frame L/f_s . It must be at least the 140-sample template length, and fine enough to separate the narrow 50 Hz line from the ECG band; but a frame spanning many heartbeats no longer contains a locally homogeneous mixture of signal and noise.

Table 3: Effect of the Wiener filter length on the complete pipeline (five K -means seeds). The true number of heartbeats is approximately 675.

L	Duration (s)	Δf (Hz)	Detected beats	Accuracy
128	0.64	1.562	844	0.4235
256	1.28	0.781	687	0.7910
512	2.56	0.391	676	1.0000
1024	5.12	0.195	675	1.0000
2048	10.24	0.098	672	1.0000

The two columns explain each other. At $L = 128$ and $L = 256$ the resolution is too coarse to isolate the mains line and the low-frequency roll-off, so the filtered signal still contains enough noise for Pan–Tompkins to over-detect — 844 and 687 peaks against approximately 675 real beats. Those spurious segments are not heartbeats at all, so no feature representation can cluster them, and a large part of the apparent accuracy loss at short L is really a segmentation failure. From

$L = 512$ the beat count is correct and the accuracy saturates. $L = 512$ is the smallest power of two that is both longer than the template and fine enough in frequency, so it is the value used. Change: $L = 256$ was previously hard-coded with no justification.

1.3.3 The desired-signal PSD model, and why the averaged template is the right one

The assignment asks whether the average PQRST complex should be treated as a template for the clean signal or for the noisy signal. It is a template for the **clean** signal: writing each beat as $x_i[t] = s[t] + n_i[t]$ with zero-mean noise independent from beat to beat, $\mathbb{E}[\bar{x}] = s$ and $\text{Var}[\bar{x}] = \sigma_n^2/N$, so averaging $N = 676$ aligned beats reduces the noise standard deviation by $\sqrt{676} \approx 26\times$ before the PSD is taken.

Rather than assert this, the notebook measures it by building a third filter in which S_d is the average of the periodograms of the *individual, un-averaged* beats — that is, by treating the template as a model of signal *plus* noise. Then $S_d \approx S_s + S_n$ and the filter degenerates to $(S_s + S_n)/(S_s + 2S_n)$, which stays within a few decibels of 0 dB everywhere and therefore passes the mains line and the baseline wander almost untouched (Fig. 3).

Table 4: Identification accuracy for three models of the desired-signal PSD, $L = 512$.

S_d estimator	Segments	Accuracy
Periodogram of the averaged template (used)	676	1.0000
Welch inside the template (first exam period)	676	1.0000
Average periodogram of the individual noisy beats	676	0.3491

At $L = 512$ the estimator used in the first exam period performs identically, because the Hann taper it applies merely smooths S_d and at this filter length the smoothing is harmless. It is nevertheless replaced, because it cannot be described correctly: with `nperseg=128` on a 140-sample input it averages a single segment.

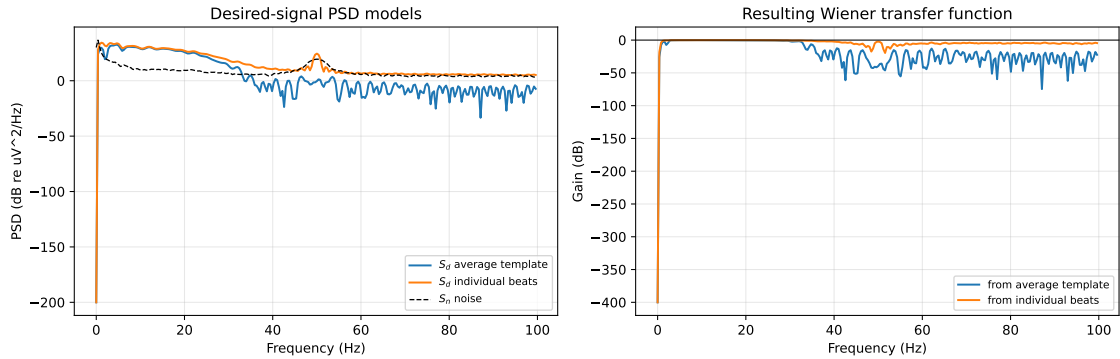


Figure 3: Left: the two candidate models of S_d against the noise PSD. Right: the resulting transfer functions. Building S_d from noisy beats yields a filter that barely filters.

1.3.4 The additional Butterworth filter was removed

The assignment permits an extra Butterworth stage *if the Wiener filtering is deemed insufficient*. With $L = 512$ it is not: the Wiener filter alone recovers the correct number of beats and perfect clustering. The 5 Hz–40 Hz band-pass used in the first exam period is also questionable in principle, since a 5 Hz high-pass removes most of the P-wave, the T-wave and the ST segment — exactly the morphological detail that distinguishes one subject from another.

Extra Butterworth	Segments	Accuracy
none (used)	676	1.0000
0.5–40 Hz	676	0.9964
1–40 Hz	676	1.0000
5–40 Hz (first exam period)	676	0.9985

Consequently the pipeline segments the Wiener output directly and the third deliverable plot (a Butterworth-filtered window with annotated R-peaks) is not required. Edge effects were checked instead: WOLA weights both analysis and synthesis with $\sqrt{\text{Hann}}$ and hops by half a frame, so the squared windows sum to a constant and reconstruction is exact in the interior. The only genuine artefact is the final incomplete frame — 174 samples, 0.87 s, 0.14 % of the record — which is removed by the boundary condition in the segmentation step.

1.3.5 Feature representation: coefficients, not sub-band energies

This is the single largest change in Part I. `pywt.wavedec(seg, 'db4', level=5)` returns `[cA5, cD5, cD4, cD3, cD2, ...]` concatenated, that is **173 coefficients** per 140-sample segment, with the sub-bands covering the octaves 0–3.1 Hz (`cA5`, P-wave and baseline) up to 50–100 Hz (`cD1`). Each coefficient is an inner product of the segment with a wavelet at a given scale *and position*, so the representation encodes how much activity of a given band occurs at a given moment in the beat.

In the first exam period each segment was reduced instead to the *energy* of each sub-band — $\sum cD3^2$ and $\sum cD4^2$ for the clustering, six energies for the PCA. Summing squares within a band discards where in the beat the energy occurs and collapses 173 numbers into six; what remains is essentially a six-bin power spectrum, so two subjects whose QRS complexes carry equal energy but differ in shape or timing become indistinguishable. It also left the dimensionality-reduction section with almost nothing to reduce.

Table 5: Effect of the feature representation, mean $\pm$ std over ten *K*-means seeds.

Representation	Dimensions	Accuracy
6 sub-band energies, <i>z</i> -scored (first exam period)	6	0.7219 ± 0.0000
6 sub-band energies, raw	6	0.3609 ± 0.0000
173 wavelet coefficients, raw	173	0.4886 ± 0.0018
173 wavelet coefficients, <i>z</i>-scored (used)	173	1.0000 ± 0.0000

Decomposition level. With 140 samples and `db4` (filter length 8) the deepest level free of boundary effects is $\lfloor \log_2(140/7) \rfloor = 4$, so PyWavelets warns at level 5. The level is prescribed by the assignment and is kept. The default `symmetric` extension applies the same boundary treatment to every segment, so the distortion is common to all of them and does not affect the *relative* comparison that the clustering is based on.

Which two coefficients are plotted. The identification algorithm may not use the labels, so the two display coefficients are selected by an unsupervised criterion: the coefficient with the largest absolute loading on PC1 and the one with the largest absolute loading on PC2 of the standardised coefficient matrix. These are the two individual features carrying most of the between-segment variation, and they are near-independent, which is what makes them useful as the axes of a scatter plot. The criterion selects `cD3[9]` and `cD4[14]` — both inside the 6–25 Hz octaves, i.e. the QRS complex, which is physiologically the most subject-specific part of the beat. The obvious alternative, largest raw variance, selects `cA5` coefficients, which track the residual baseline level and separate recording sessions rather than morphology.

1.3.6 K-means settings and run-to-run stability

$K = 5$ is given by the problem. `n_init=10` and `random_state=42` are fixed so the notebook is reproducible. Lloyd’s algorithm converges to a *local* minimum whose identity depends on the random initialisation, so the assignment’s question about reproducibility is answered by measurement rather than assertion: over ten seeds the standard deviation of the accuracy is exactly 0 in the full coefficient space and for every $N \geq 6$ after PCA, because the five subjects form well-separated compact groups there. Visible seed-to-seed variation appears only in the strongly compressed representations ($N = 2$: 0.8466 ± 0.0019), which is the situation the question points at.

Cluster *labels* are arbitrary — any permutation of the five indices describes the same partition — so accuracy is only meaningful after the best of the $5! = 120$ permutations has been applied, which is what `match_labels` does.

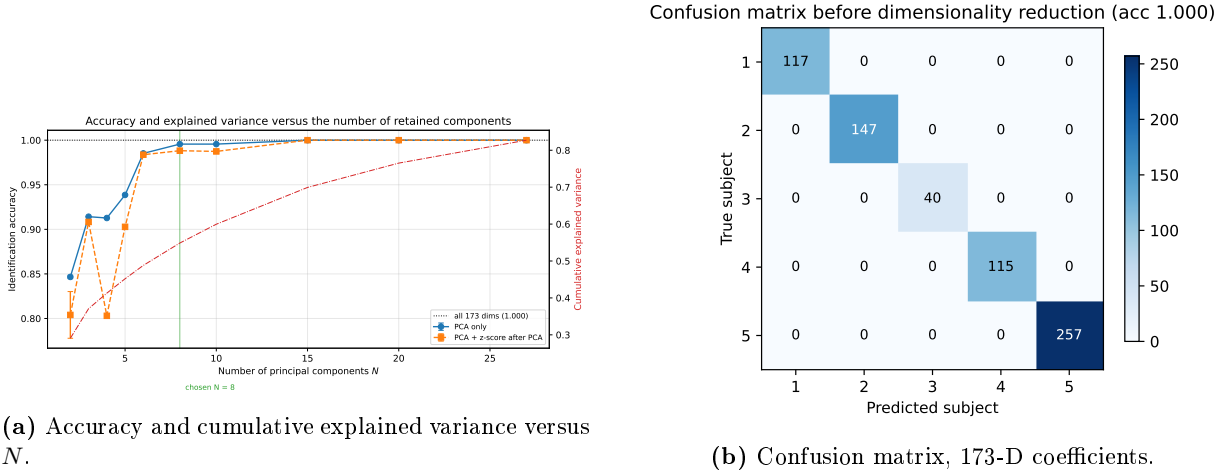


Figure 4: Dimensionality reduction. The accuracy curve saturates around $N = 8$, at which point only 55 % of the variance is retained — the 90 % rule would keep 40 components.

1.4 Results

Clustering the 676 segments in the standardised 173-dimensional wavelet space recovers the five subjects **exactly** (accuracy 1.000; the confusion matrix is diagonal, with 117, 147, 40, 115 and 257 segments correctly assigned). After PCA to eight dimensions the accuracy is 0.9956, with three segments of subject 1 assigned to subject 2.

A caveat on the perfect score. This is a small, controlled problem: five subjects, one sensor, one recording session each, and one of the five (subject 3) has valvular disease, which makes their beats morphologically unmistakable. It should not be read as evidence that ECG biometrics generalise at this level — with more subjects, recordings on different days, or different electrode positions, within-subject variability would grow and the between-subject margins would shrink. Note also that subject 3 contributes only 5.9 % of the segments, so a degenerate clustering that merged that subject into a neighbour would still score 94 %; this is why the confusion matrix, and not only the scalar accuracy, is reported.

2 Part II — EEG-based sleep staging

2.1 Pipeline

Twenty overnight PSG recordings, four channels (Fpz-Cz, Pz-Oz, EOG, EMG) at 100 Hz, segmented into 30 s epochs: 21 357 epochs in total, 15 760 from the training subjects 0–14 and 5597 from the held-out subjects 15–19. The chain is: FastICA artefact removal, 0.5–30 Hz band-pass,

Welch PSD and band-power features, leave-one-subject-out model selection, and finally retraining on all training subjects with a single evaluation on the held-out subjects.

2.2 Answers to the overview-sheet questions

What visible patterns characterise each sleep stage?

The five epochs discussed below all come from subject 6; epoch indices are given for each.

- **Wake (epoch 5).** Fast, irregular, relatively high-amplitude EEG. Physiologically wakefulness shows low-amplitude beta, with alpha (8–12 Hz) when the eyes are closed and the subject is relaxed; much of the visible amplitude here is in fact ocular and muscular contamination, which is what the ICA step removes. The supporting evidence is in the reference channels: frequent large EOG deflections and the *highest* EMG level of the five epochs, reflecting active postural muscle tone.
- **N1 (epoch 64).** The EEG slows and its amplitude falls; alpha drops out and theta (4–8 Hz) dominates. N1 has no positive defining marker — it is recognised by the *absence* of both alpha and the N2 markers — which is what makes it the hardest stage to score. The EOG shows slow rolling deflections rather than the sharp movements of wake.
- **N2 (epoch 74).** Medium-amplitude theta background interrupted by **sleep spindles**: 11–16 Hz bursts lasting about a second, one visible near $t = 25$ s. K-complexes (a sharp negative deflection followed by a positive component) may also occur. The EOG is quiet and the EMG is further reduced.
- **N3 (epoch 173).** Large-amplitude, slow, strongly synchronised delta activity below 4 Hz; the trace is markedly more regular than in the lighter stages. Ocular contamination is still visible in the EEG channels even here, which is why artefact removal precedes every feature computation.
- **REM (epoch 255).** A low-amplitude, mixed-frequency EEG that resembles wake — hence “paradoxical” sleep — but the reference channels resolve the ambiguity: distinct *rapid* EOG deflections, and the *lowest* EMG level of all five epochs, reflecting REM muscle atonia.

The last point drives a modelling decision. Wake and REM are nearly indistinguishable on the EEG channel alone and are separated almost entirely by the EOG and EMG channels, which is why those channels are evaluated as *features* in Task 4 and not only as artefact references.

What is the effect of including versus excluding EOG/EMG channels from ICA input?

Including them is what makes the problem solvable, for two distinct reasons.

1. **Identifiability.** ICA can separate at most as many sources as it has input channels. With four inputs it resolves four sources; with the two EEG derivations alone it could resolve two, fewer than the number of physically present sources (cortical activity on two derivations, ocular, muscular). The ocular and muscular contributions would then necessarily remain mixed into the EEG components.
2. **Labelling.** Even if a two-channel ICA did isolate an artefact, nothing would say *which* component it is. The automatic rule needs a reference signal to correlate against, and the EOG and EMG channels are exactly that. This is not merely convenient here: as §2.3.3 shows, the components cannot be told apart by their spectra in this dataset, so the reference channels are necessary.

The cost is that two of four components are discarded, so the reconstruction is rank-deficient and whatever genuine cortical activity projected onto them is lost with the artefacts. That is the standard trade-off of ICA-based artefact rejection, and it is one reason the EOG and EMG channels are re-introduced as separate features rather than discarded entirely.

Should filtering be done before or after ICA artefact removal? Why?

After — ICA first, band-pass second, which is the order used.

ICA models each observation as an *instantaneous linear mixture* $x[t] = A s[t]$ and estimates a single mixing matrix A from the joint amplitude distribution of the four channels. Filtering first breaks that model: a band-pass is applied per channel and affects the underlying sources differently depending on where their energy lies, so the observed data are no longer an instantaneous mixture with one constant matrix. Filtering first would also remove the evidence the identification step needs — the artefacts are characterised precisely by the frequency ranges a 0.5–30 Hz band-pass attenuates, so the correlations against the reference channels would weaken. The reverse argument does not apply: nothing about the band-pass depends on the artefacts having been removed, so there is no cost to placing it second.

A caveat worth stating: a very large drift can degrade the ICA estimate itself, and the usual remedy is a gentle high-pass well below the band of interest *before* ICA with the full band-pass afterwards. That was not needed here — the components separate cleanly, with $|r| > 0.997$ against the reference channels for all twenty subjects.

Describe the pipeline that worked best, including preprocessing, model and hyperparameters. Why is this the best model?

Features	Welch PSD of Fpz-Cz, 0.5–30 Hz, 3 s segments, in dB (89 bins) plus 5 EOG/EMG features (94 in total)
Normalisation	z-score within each recording, then a global <code>StandardScaler</code>
Model	SVC, RBF kernel
Hyperparameters	<code>C = 1.0, gamma='scale', class_weight='balanced'</code>
LOSO accuracy	0.780 ± 0.098
LOSO F1-macro	0.727 ± 0.101
LOSO F1-weighted	0.794 ± 0.091

Three observations drive the choice.

The preprocessing matters more than the classifier. Averaged over classifier families, F1-macro ranges from 0.598 to 0.702 across the seven preprocessing configurations, whereas within the plain PSD block the nine classifier settings span only 0.578 to 0.671 (Table 6). The largest single gain is per-recording normalisation. The reason is physical rather than statistical: absolute EEG amplitude depends on skull thickness, electrode impedance, age and amplifier gain, so different nights sit at different absolute dB levels, and those offsets are larger than the between-stage differences the classifier must detect. A global `StandardScaler` cannot remove them because it subtracts the *grand* mean over all training epochs, not each recording’s own offset. Normalising within a recording turns the features into “deviation from this person’s average night”.

The EOG/EMG features fix the stages the EEG cannot separate. Adding five features derived from the reference channels raises the best F1-macro from 0.675 to 0.727. Wake and REM have similar EEG spectra; the EMG envelope separates them almost by itself (Cohen’s $d = 7.10$ between Wake and REM on subject 6).

Among classifiers, the RBF SVM is strongest. With 94 standardised features and about 15000 training epochs, an RBF kernel can represent the curved boundaries between neighbouring stages, and the margin formulation is comparatively robust in a moderately high-dimensional space where k -NN distances become uninformative and single trees find unstable splits. Logistic regression is close behind (0.722), which suggests the classes are close to linearly separable once the features are right — another sign that the representation, not the model, is doing the work. `class_weight='balanced'` trades about two points of accuracy for a larger gain in F1-macro, which is the intended trade given the selection criterion (on the plain PSD block: $0.654 \rightarrow 0.671$ F1-macro while accuracy falls $0.783 \rightarrow 0.755$).

Table 6: Mean and best F1-macro per preprocessing configuration across the classifier families.

Preprocessing	Features	Mean F1-macro	Best F1-macro
PSD plus EOG/EMG, per-recording z -score (used)	94	0.702	0.727
PSD, per-recording z -score	89	0.650	0.675
PSD, global scaling only	89	0.632	0.671
PSD then PCA, 10 components	10	0.629	0.672
$\log_{10}$ band power, 2 channels	10	0.623	0.655
Relative band power, 2 channels	10	0.616	0.641
dB PSD averaged in the 5 bands, 2 channels	10	0.598	0.650

How does the model perform on the held-out test set, and how does that compare to the cross-validation performance?

On subjects 15–19 (5597 unseen epochs) the deployed pipeline reaches **accuracy 0.780**, **F1-macro 0.719** and F1-weighted 0.794.

Table 7: Classification report on the held-out subjects, with the first exam period for comparison.

Stage	Precision	Recall	F1	Support	F1, first exam period
Wake	0.749	0.760	0.754	890	0.728
N1	0.269	0.621	0.375	264	0.062
N2	0.891	0.792	0.838	2351	0.817
N3	0.813	0.827	0.820	958	0.800
REM	0.853	0.768	0.808	1134	0.641
Accuracy		0.780		5597	0.744
Macro avg	0.715	0.753	0.719	5597	0.610
Weighted avg	0.818	0.780	0.794	5597	0.729

The test result is very close to the cross-validated estimate — F1-macro 0.719 against 0.727, accuracy 0.780 against 0.780 — which is the useful outcome: the LOSO protocol, computed only on subjects 0–14, predicted performance on five subjects the model had never seen. The small drop is what one expects when moving from a validation estimate to a single held-out evaluation, and it does not indicate overfitting.

The pooled number should be read alongside the per-subject breakdown, because the spread across nights is much wider than the gap between the CV and test means:

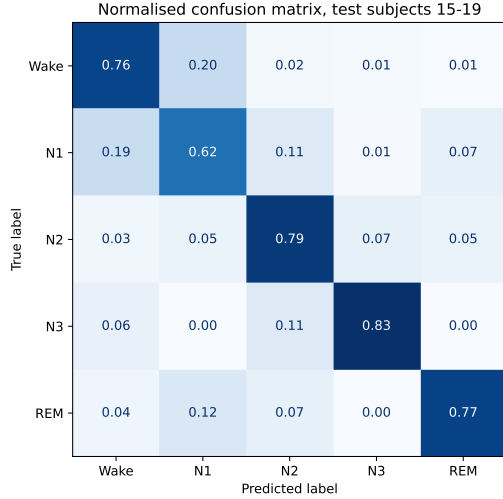
Subject	15	16	17	18	19
Epochs	952	1144	1002	964	1535
Accuracy	0.871	0.742	0.734	0.771	0.787
F1-macro	0.814	0.690	0.629	0.716	0.722

Performance depends more on *which* subject is being scored than on the train/test split, which is also why the LOSO fold standard deviation (± 0.101 on F1-macro) is large.

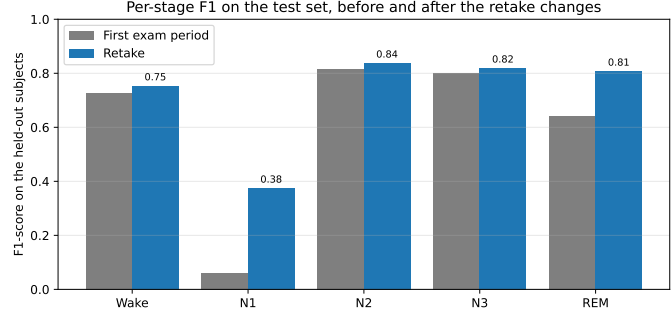
2.3 Design choices not covered by the overview sheet

2.3.1 ICA: fitting data, reshaping, and how much data is needed

`FastICA.fit` expects (`n_samples`, `n_features`), where a feature is a recording channel and a sample is one instant in time. The raw array is (`n_epochs`, `n_channels`, `n_samples`), so time and channels are swapped and the first two axes are collapsed, giving (`n_epochs` $\times$ `n_samples`, 4). Collapsing epochs is legitimate because ICA has no notion of time order — it models each observation as an instantaneous mixture — so the discontinuities at epoch boundaries are irrelevant.



(a) Normalised confusion matrix, subjects 15–19.



(b) Per-stage F1 before and after the retake changes.

Figure 5: Test-set performance. Every stage improved; N1 rose from F1 0.062 to 0.375 and REM from 0.641 to 0.808.

The model is fitted on the **training subjects only** (0–14) and then applied unchanged to the test subjects, so no information from subjects 15–19 enters the preprocessing. `random_state` is fixed, because FastICA starts from a random unmixing matrix and neither the sign nor the order of the recovered components is identified by the model.

Change: the fit now uses a seeded random subsample of 100 000 samples per subject (1500 000 by 4) instead of all 47 280 000 by 4 samples. The quantity being estimated is a single 4×4 unmixing matrix — sixteen parameters — and FastICA converges long before millions of samples; the full fit cost minutes and bought nothing. The notebook verifies the claim by re-estimating on a four times larger disjoint subsample and comparing the mixing directions after removing the sign and permutation ambiguity: the best cosine similarities are [1.000, 0.9999, 1.000, 1.000].

2.3.2 Automatic component selection

Each component’s concatenated time series is correlated with the EOG and EMG reference channels of the same subject, and the component with maximum $|r|$ against each reference is zeroed before the inverse transform. The selection is recomputed per subject, so the procedure adapts to per-recording differences in the mixing.

Change: the selection is now audited. The notebook prints, for all twenty subjects, which component was chosen and how strong the association was, and flags the case where the same component would be nominated for both references (it never occurs). The correlations are extremely high — $|r| \geq 0.998$ for every subject — and the reason is structural: with four input channels and four components, two of which *are* the reference signals, ICA can place the ocular and muscular sources almost exactly onto the EOG and EMG channels themselves, so the recovered components are near-copies of those channels rather than blends. On the example epoch, removing them reduces the Fpz-Cz standard deviation from $183.9 \mu\text{V}$ to $27.5 \mu\text{V}$.

Two implementation notes. The transform is applied in blocks of 200 epochs, which bounds peak memory by a constant rather than by the length of the recording; and the correlations are accumulated across blocks with a numerically stable pairwise co-moment update in `float64`. The naive route — accumulating raw sums of squares in the `float32` dtype of the recordings — suffers catastrophic cancellation over millions of samples and returns $|r|$ slightly *above* 1, which is how the problem was noticed.

2.3.3 The EMG channel is an envelope, not raw EMG

This deserves its own subsection because it invalidates the textbook reasoning and changed two decisions. Measured over all epochs of subject 6:

Channel	Std (μ V)	% power <4 Hz	4–15 Hz	>15 Hz	Distinct values per epoch
Fpz-Cz	85.03	95.57	3.35	1.08	1287
Pz-Oz	59.85	96.55	2.66	0.80	1257
EOG	67.33	96.81	2.46	0.73	917
EMG	1.04	99.99	0.01	0.01	27

Twenty-seven distinct values across 3000 samples, a standard deviation two orders of magnitude below the EEG, and essentially no power above 4 Hz: this channel is a submental *envelope* sampled at 1 Hz and zero-order held to 100 Hz, not a raw EMG trace. Raw surface EMG is the summation of short motor-unit action potentials and is therefore broadband above 20 Hz; that signature is absent here by construction.

Two consequences. First, the artefact components cannot be distinguished by their spectra in this dataset — both the ocular and the muscular component sit at low frequency — so the correlation-based rule of §2.3.2 is not merely convenient but necessary. Second, the textbook way to quantify muscle tone, a high-frequency EMG band power, would measure an empty band; the envelope’s *level* is what encodes tone, so the mean and the standard deviation of the envelope are used instead.

Correction. The first exam period identified the EMG component on the grounds that it “represents a very slow wave, as we have in the EMG signal”. The reasoning was wrong — EMG is physiologically the fastest of the four signals — but the observation about this particular recording was right, for a reason that had not been identified. My own first draft of this retake made the opposite error, using a 20–45 Hz EMG band power on the strength of the physiology; the channel table above is what caught it.

2.3.4 Filter design

Butterworth, order 4, 0.5–30 Hz, applied as second-order sections with `sosfiltfilt`. Butterworth because it is maximally flat in the pass-band and therefore does not impose ripple on the very spectral shape the PSD features measure; second-order sections for numerical stability; `sosfiltfilt` because filtering forwards and backwards cancels the phase response, and sleep-scoring markers such as K-complexes are defined by their waveform shape, which a non-linear-phase filter would distort (the price is a squared magnitude response, i.e. an effective 8th order with -6 dB rather than -3 dB at the corners). The pass-band covers every band used for staging — delta, theta, alpha, sigma/spindles and beta — so no discriminative information is discarded, while the high-pass removes electrode drift and the low-pass removes residual noise. Measured attenuation: -7.4 dB below 0.5 Hz and -16.9 dB above 30 Hz.

Change: in the first exam period this filter was defined and applied to a single epoch for the demonstration plot, while feature extraction read the ICA output directly. Task 3 asks for it to be applied to all subjects, and the written answer to Task 4 argued that filtering is necessary — but the pipeline never did it. It is now applied to every epoch of every subject.

Why filter at all, if the out-of-band PSD bins are discarded afterwards? Because discarding out-of-band *bins* is not the same as removing out-of-band *power*. Welch multiplies each 3 s segment by a Hann window whose frequency response has sidelobes of finite height, so any strong component outside the band of interest deposits energy inside the retained bins through those sidelobes. Here the two offenders are large: electrode drift below 0.5 Hz, many dB above the delta band, and broadband noise above 30 Hz. Since the features are $10 \log_{10}$ of each bin, that contamination enters the feature vector directly. Filtering removes the offending power before it can leak.

2.3.5 Feature extraction

Welch with `nperseg=300` (3 s segments, $\Delta f = 0.333 \text{ Hz}$), converted to dB. Change: only the 89 bins in 0.5–30 Hz are kept, instead of all 151 bins up to 50 Hz. The bins above 30 Hz lie in the stop-band of the filter, so they contain the roll-off and the numerical floor rather than physiology — and that floor is subject-specific (electrode impedance, amplifier gain), which is precisely the kind of feature that helps within a subject and hurts across subjects under leave-one-subject-out.

Change: band power is never used raw. It is a positive, strongly skewed quantity spanning orders of magnitude, so it is entered either as $\log_{10}$ (variance-stabilising, turning multiplicative differences into additive ones — the same reason the PSD features are in dB) or as relative power normalised by the epoch total (which removes the epoch's overall amplitude and leaves the spectral shape that defines a stage). In the first exam period the absolute linear power in volts (values around $1\text{e-}10$) was fed to the classifiers directly, which is why every band-power pipeline lost to the raw PSD.

Five EOG/EMG features are added: EOG power in 0.5–4 Hz (slow rolling movements) and 4–15 Hz (rapid movements), EOG amplitude, and the mean level and standard deviation of the EMG envelope, for the reasons in §2.3.3.

2.3.6 Cross-validation protocol

Change: the split is now `LeaveOneGroupOut` with an explicit `groups` array of subject ids, passed to `cross_validate` with a scoring dictionary and `n_jobs=2` — both are named deliverables and neither was used in the first exam period, which implemented the loop by hand and therefore ran serially for 3.5 hours.

Why the folds must be subject-separated at all: epochs from one night are not independent samples — they share an electrode montage, skull anatomy, age and medication, and neighbouring epochs are temporally correlated. A plain k -fold would place epochs of the same night in both the training and the validation set, letting the model recognise the subject instead of the sleep stage and producing an optimistic score. LOSO reproduces the deployment situation: a completely new patient.

One deliberate deviation, stated explicitly. Each fold *trains* on every second epoch of the fourteen training subjects, while *validation* always uses every epoch of the held-out subject. The reported metrics are therefore computed on complete recordings and remain unbiased; only the amount of training data used during model *selection* is halved, which cut the cost of the kernel and ensemble models by roughly a factor of three and made the grid feasible on the two-core machine this was run on. The selected pipeline is retrained on the *full* training set before the test evaluation. Subsampling for selection and refitting on everything is standard practice; the risk it carries is that the ranking could differ slightly at full sample size, which is why the top of the table (0.727 for the SVM against 0.722 for logistic regression, both on the same feature block) is treated as a narrow margin rather than a decisive one.

Grid. Change: the grid is organised as a screen plus a targeted sweep instead of a full outer product. The cached results of the first exam period — reproduced in the notebook — show the ranking there was driven almost entirely by the preprocessing rather than by the classifier hyperparameters, so here all seven preprocessing configurations are screened with one representative setting per classifier family, and the hyperparameter variation the assignment asks for (k for k -NN, `class_weight` and C for the linear and kernel models) is run on the plain PSD block. That is **39 pipelines $\times$ 15 folds = 585 model fits**, all of which were actually run; nothing appears in the table that was not measured. `random_state` is set on every stochastic estimator — in the first exam period the random forest and the decision tree had none, so that table was not reproducible despite a global seed at the top of the notebook.

Metrics. Accuracy, F1-macro and F1-weighted, with **F1-macro as the selection criterion** because the classes are strongly imbalanced (N2 about 40 % of epochs, N1 under 6 %) and a classifier that never predicts N1 loses under six points of accuracy while being useless for detecting

fragmented sleep. Selecting on a metric that treats classes equally is also what motivates including `class_weight='balanced'` in the grid: it would be inconsistent to select on such a metric while giving every candidate a loss function that weights samples equally.

2.3.7 Validity of the per-recording normalisation

Per-recording z -scoring is *transductive*: the statistics of a test subject are computed from that test subject’s own epochs. This uses no labels, and offline PSG scoring always has the complete recording available before staging begins, so it does not use information that would be unavailable in deployment. It is stated here rather than hidden because it does impose a constraint: the method could not be used for real-time staging, where a causal alternative — normalising against a running estimate from the epochs seen so far — would be required.

2.4 Are the results in line with expectations, and what would improve them?

Broadly yes, and the pattern of errors is the physiologically expected one. The model is strongest on the stages with distinctive spectral signatures (N3, dominated by high-amplitude delta; N2, the most frequent stage and marked by sigma-band spindles). Wake is recovered well once the EMG feature is available, and REM improves substantially for the same reason in reverse.

N1 remains the weakest class at F1 0.375, and that is expected rather than a defect of this particular model: N1 is a transition state defined largely by what is *absent*, it is the rarest stage in the dataset, and inter-rater agreement between human scorers on N1 is the lowest of the five stages. A model trained on a single frontal derivation cannot be expected to exceed the agreement of the experts who produced its labels. Note the shape of the improvement: recall rose from 0.034 to 0.621 while precision stayed low at 0.269, i.e. `class_weight='balanced'` converted a classifier that ignored N1 into one that over-predicts it. Whether that is the better failure mode depends on the application; for detecting fragmented sleep it is.

Three directions would improve the result further, in order of expected benefit.

1. **Use temporal context.** Every epoch is currently classified in isolation, yet sleep has strong sequential structure: N3 does not follow wake directly, and stages persist for many epochs. The isolated single-epoch flips visible in the hypnogram (Fig. 6) are exactly the errors this would remove. The cheapest version appends the features of neighbouring epochs to each feature vector; a principled version models the stage sequence with a hidden Markov model or a conditional random field whose transition matrix is estimated from the training hypnograms, decoding over the whole night rather than epoch by epoch.
2. **Target N1 directly.** The underlying difficulty is that N1 has no positive marker. Features describing the *transition* — the rate of change of alpha power between consecutive epochs, slow rolling eye movements in the EOG — would give the classifier something to detect rather than an absence to infer.
3. **More subjects.** With fifteen training nights the LOSO fold spread is wide, which means the model is limited as much by how few subjects it has seen as by its capacity. Additional recordings would help more than further hyperparameter tuning.

3 Changes with respect to the first exam period

Each row corresponds to one or more `# RETAKE CHANGE` comments in the code.

What	First exam period	Retake	Ref.
Part I			
Wavelet features	6 sub-band energies; clustering on 2 of them	173 concatenated db4 coefficients, z -scored	§1.3.5

What	First exam period	Retake	Ref.
Wiener filter length	$L = 256$, assumed	$L = 512$, chosen by a sweep	§1.3.2
S_d estimator	Welch on a 140-sample template (averages one segment)	Periodogram of the zero-padded template	§1.3.3
Extra Butterworth	5–40 Hz band-pass	Removed; Wiener alone suffices	§1.3.4
dB convention	$20 \log_{10}$ on the PSD plots	$10 \log_{10}$ for PSDs, $20 \log_{10}$ for $ W $	Fig. 2
PCA components	3, from a 90 % variance rule on 6 dims	8, from an accuracy sweep on 173 dims	Fig. 4a
Display coefficients	Asserted	Chosen by PC1/PC2 loadings	§1.3.5
Detector validation	None	Beat count estimated from the label track	Tab. 2
Reproducibility	Unseeded <code>random.randint</code>	Seeded generator	—
New figures	—	Raw PSD, L sweep, S_d comparison, PCA sweep, edge effects, subject overlay	—
Part II			
Band-pass filter	Applied to one epoch only	Applied to every epoch of every subject	§2.3.4
PSD feature range	151 bins to 50 Hz	89 bins, 0.5–30 Hz	§2.3.5
Band power	Absolute linear power, raw	$\log_{10}$ and relative power	§2.3.5
Normalisation	Global <code>StandardScaler</code> only	Per-recording z -score added	Tab. 6
EOG/EMG channels	Excluded as features	Five features added	§2.3.3
EMG rationale	“EMG is a very slow wave”	Channel shown to be a 1 Hz envelope; features chosen accordingly	§2.3.3
Cross-validation	Hand-written loop, 3.5 h	<code>LeaveOneGroupOut</code> plus <code>cross_validate</code> , <code>n_jobs=2</code>	§2.3.6
Class imbalance	Not addressed	<code>class_weight='balanced'</code> in the grid	§2.3.6
Reproducibility	No <code>random_state</code> on RF/DT	Set on every stochastic estimator	§2.3.6
ICA fit	All 47 280 000 samples	Seeded 1 500 000-sample subsample, verified	§2.3.1
Component selection	Not audited	Per-subject table; stable streaming correlation	§2.3.2
Units	Volts internally, μV on axis labels	Converted to μV once at load	—
Memory and runtime	<code>X</code> and <code>X_clean_all</code> held for all subjects	Streamed per subject; caching	—
New figures	—	Channel PSD, IC selection, per-class F1, hypnogram, EOG/EMG box plots	—

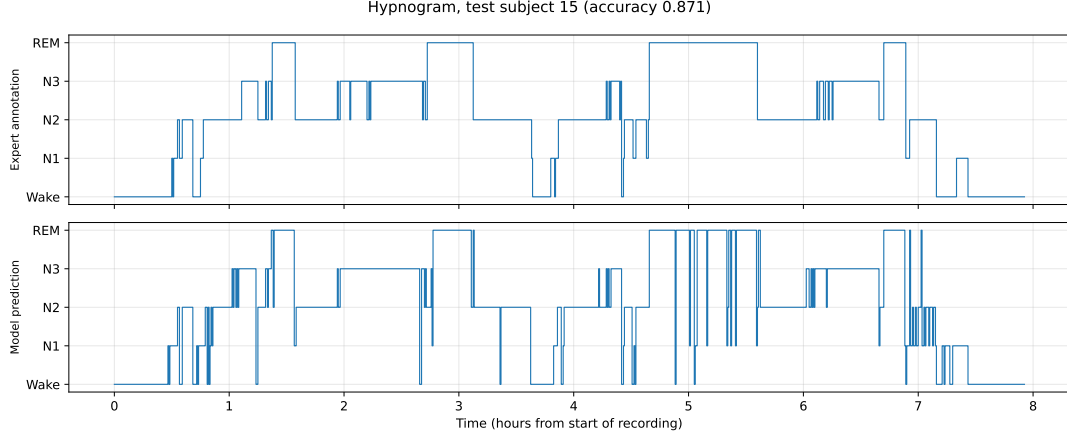


Figure 6: Expert annotation and model prediction for test subject 15 (accuracy 0.871). The architecture of the night is reproduced; residual errors concentrate at stage transitions and in isolated single-epoch flips.

Summary of the effect. Part I: identification accuracy 0.558 to 1.000 without dimensionality reduction, and 0.773 to 0.996 with it. Part II: LOSO F1-macro 0.678 to 0.727; on the held-out subjects, accuracy 0.744 to 0.780 and F1-macro 0.610 to 0.719, with every individual stage improving and N1 rising from F1 0.062 to 0.375.